5 (a) (i)	force per unit positive charge (ratio idea essential)		
(ii)	$E = Q / 4\pi\varepsilon_0 r^2$	M1	
	ε_0 being the permittivity of free space	A1	
(b) (i)	$2.0 \times 10^6 = Q / (4\pi \times 8.85 \times 10^{-12} \times 0.35^2)$	C1	
	$Q = 2.7 \times 10^{-5}$ C	A1	**[2]**
(ii)	$V = (2.7 \times 10^{-5}) / (4\pi \times 8.85 \times 10^{-12} \times 0.35)$	C1	
	$= 7.0 \times 10^5$ V	A1	**[2]**
(c)	electrons are stripped off the atoms	B1	
	electrons and positive ions move in opposite directions, (giving rise to a current)	B1	**[2]**
6 (a) (i)	arrow B in correct direction (down the page)	B1	
(ii)	arrow F in correct direction (towards Y)	B1	**[2]**
(b) (i)	When two bodies interact, force on one body is equal but opposite in direction to force on the other body.	B1	**[1]**
(ii)	direction opposite to that in **(a)(ii)**	B1	**[1]**
(c)	suggested reasonable values of I and d	B1	
	mention of expression $F = BIL$	B1	
	force between wires is small	M1	
	compared to weight of wire	A1	**[4]**
7 (a)	'uniform' distribution	B1	**[1]**